

Total Points

100 / 110 pts

Question 1

Q1

10 / 10 pts

✓ - 0 pts Correct

Question 2

Q2

0 / 10 pts

✓ - 10 pts Incorrect or vague explanation

Question 3

Q3

10 / 10 pts

✓ - 0 pts Correct

Question 4

Q4

10 / 10 pts

✓ - 0 pts Correct

Question 5

Q5

10 / 10 pts

✓ - 0 pts Correct

2^{24} Bytes

Question 6

Q6

10 / 10 pts

✓ - 0 pts Correct

Question 7

Q7

10 / 10 pts

✓ - 0 pts Correct

Question 8

Q8

10 / 10 pts

✓ - 0 pts Correct

Question 9

Q9

10 / 10 pts

✓ - 0 pts Correct (needed three)

Question 10

Q10

10 / 10 pts

✓ - 0 pts Correct

Question 11

Q11

10 / 10 pts

✓ - 0 pts Correct

1. What is deadlock?

A deadlock is a synchronization problem where processes / threads are waiting on each other for a shared resource leading to a perpetual wait. More formally, in a set of processes / threads, each process / thread is waiting on a shared resource held by another process / thread in the same set, leading to perpetual wait.

2. Explain why in the Dining Philosophers problem picking up chopsticks inside a critical section ("picking up chopsticks simultaneously") isn't efficient?

The Dining Philosophers problem is a classic synchronization problem. Picking up the chopsticks simultaneously is not efficient as each philosopher ~~will~~ attempt to pick up both chopsticks at the ~~same~~ time, but if they fail to pick up one, they will have to wait for a chopstick to be released by another philosopher. However, this can lead to a synchronization problem called deadlock. Suppose all philosophers pick up their left chopstick. Each one is waiting for a right chopstick to be released but the philosopher holding these are also waiting. This will lead to perpetual waiting.

3. Describe the "Readers-Writers" problem. No solution is required. Just formulate the problem.

The ~~Readers-Writers~~ Readers-Writers problem is a classic synchronization problem. There are reader processes/threads that can access and read a shared resource and there are writer processes/threads that can mutate a shared resource. Issues arise when writers try to mutate a shared resource at the same time, such as race condition. In addition, if a reader tries to read the resource while a writer is mutating it, it might incorrectly read stale data.

4. What is memory management unit (MMU)? Be sure to mention its main function.

MMU is a memory device that converts logical addresses to physical addresses.

5. How much memory can the following system have?

- Page size 4 kB
- CPU frequency 2 GHz
- Logical and physical address size 24 bits
- TLB size 64 entries
- Memory is byte-addressable (most common one and likely the only one you know)

2^{24} bytes of memory/addresses. This is because 24 bits can only represent 2^{24} numbers and we are given that the address size is 24 bits.

6. Briefly discuss advantages and disadvantages of using shared memory for IPC.

Shared memory is a type of IPC where multiple processes share the same memory address space to communicate information. Some advantages it has over message passing, the other type of IPC, is that it is faster because it doesn't have the overhead of using the OS as a middle man like in message passing. Some disadvantages are that race conditions may occur when multiple processes try to access a shared resource.

7. Why IP address isn't enough for network communications? What is socket? For this question consider only Internet sockets that we discussed in class.

An IP address is a unique identifier for a computer on a network (usually a number). It is not enough for network communications as the networking port is required to identify the processes that is involved in the communication. A networking port is a unique identifier for a process on a computer that wants to use the network. A socket is the combination of an IP Address and networking port.

8. What is URL? In this question, I don't care what URL stands for. Just explain me the idea of this thing.

A URL is a human-readable identifier for a web resource on a network. It can be converted to an IP address through the DNS server.

9. What are the advantages of threads compared to processes?

Threads are faster to delete/create. They also have implicit shared memory communication. In addition, multiple threads can be run on one CPU core while multiple processes cannot.

10. What is stack of a process (a.k.a. runtime stack)? What is stored there?

The stack is a piece of memory allocated for a process/thread. Local variables, function calls, and return addresses of a process/thread are stored there.

11. There is the following system.

Logical address length: 24 bits

Physical address length: 24 bits

Page size: 4 kB

$$\rightarrow 4 \times 1024 = 2^2 \times 2^{10} = 2^{12} \rightarrow 12 \text{ bits}$$

TLB buffer looks like this

Page number	Frame number
0	9
1	10
2	11
3	12
4	13
5	14
6	15
7	16

$$\begin{array}{r} 2^{11} \\ - 12 \\ \hline 12 \\ 2^{12} \text{ pages} \end{array}$$

A) what is the size of page offset (in bits): 12 bits

B) what is the number of entries in a page table: $2^{12} = 4096$ entries

C) What physical address corresponds to the following logical address?

The answer should be in hexadecimal form. 0x00F7E4

